

agentLUMEN

AGENTIC FLASHLIGHT CO

**Agents handle routine operations
Humans govern strategy, exceptions, and
high-stakes decisions**

Build vs Buy Matrix

BUY (SaaS Tools)

Stripe/
Revolut

ShipStn

Zendesk

DataDog /
PowerBI

DataDog /
PowerBI

Xero

Commodity | PCI Handled | Fast

BUILD (Custom Moat)

Orchestration
(RLM)

Context Graph

Transaction
Firewall

Custom
Agents

Pricing Engine

Security
Observer

IP Moat | Decision Trace| Control

SaaS:

**Tools behind a gateway
(not autonomous agents)**

BUSINESS ARCHITECTURAL DECISIONS

AGENTIC DECISIONS (WHY IT WORKS)

Multi-agent, single orchestrator

- Specialists coordinated centrally
- No agent-to-agent calls (limits blast)

Commerce-grade autonomy (not a chatbot)

- Availability → checkout → pay → refund
- Transaction firewall + idempotency

Auditability by design

- Bi-temporal logs + decision provenance
- Security Observer is read-only evidence

PHYSICAL DECISIONS (WHY IT'S DEPLOYABLE)

Hybrid: Colo + Cloud Burst

- PII + sensitive IP stay in colo
- Burst compute to cloud for elasticity/cost

Security + governance as built-in primitives

- Zero-trust posture (assume prompt injection)
- Policy-as-code (ABAC), step-up auth, HIL>\$250

Compliance-ready evidence

- ISO 27001 / ISO 42001 / EU AI Act / PCI DSS
- Append-only decision trail for audits

Bottom line: agentLumen is a safe, auditable commerce agent architecture that scales via cloud without sacrificing sovereignty, IP protection, or payment-grade controls.

Logical $\Rightarrow$ Physical Mapping

Business Driver

- Protect IP + secrets
- Audit trail for compliance
- Prevent AI misuse
- Elastic customer traffic
- Reduce PCI scope
- Cost efficiency

Architecture Decision

- Custom agents (not SaaS)
- Bi-temporal context graph
- Transaction Firewall
- Managed storefront + CDN
- SaaS payments (Stripe)
- Hybrid 70/30 split

Physical Placement

- COLO
- COLO (PII zone)
- COLO (isolated)
- CLOUD
- EXTERNAL
- COLO + CLOUD

BUILD (IP Moat)

- Orchestrator (RLM)
- Domain Agents
- Transaction Firewall
- Security Observer

DEPLOY (Managed)

- Neo4j, Qdrant
- PostgreSQL, Redis
- API Gateway

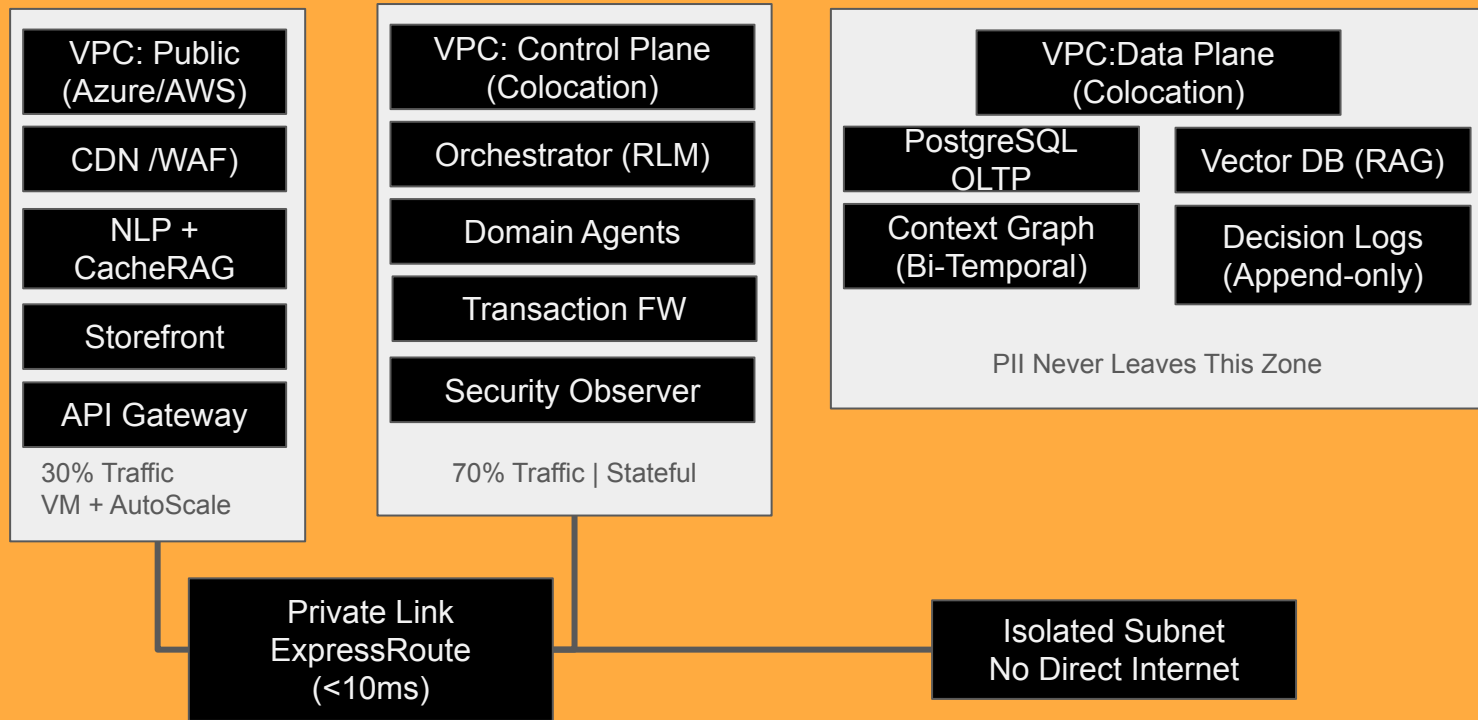
BUILD (IP Moat)

- Stripe, ShipStation
- Zendesk, Xero, Intercom
- DataDog, Cloudflare

TOGAF

Business Architecture $\rightarrow$ Data/App/Tech Architecture $\rightarrow$ Implementation

Hybrid Deployment + Network Segmentation



Data Placement: Where everything lives

COLOCATION Data Plane

PostgreSQL (OLTP)

- Orders, Customers
- Inventory, Products

Context Graph (Neo4j)

- Bi-temporal trace
- Decision provenance

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PII + Sensitive

COLOCATION Control Plane

Redis (Session 3h)

QDrant / Pinecone (Vec)
Neo4j (Context Graph)

Agent Runtime (GPU)

Orchestrator
Transactional FW

Security Observer

Compute + Memory

CLOUD (AZURE / AWS) Analytics

BigQuery (OLAP)

- Aggregated only
- No PII
- RAGAS eval metrics

DataDog (SIEM/Monitoring)

- Logs Alerts
- Redacted Traces

PowerBI (Dashboards)

- Business KPIs

Analytics + Monitoring

RULE:

PII never leaves colo · Analytics gets aggregates only · Logs are redacted

SECURITY + COMPLIANCE ARCHITECTURE

ZERO-TRUST AGENT MODEL

"Every agent assumed compromised"

- Microsegmented VPCs
- No agent-to-agent direct calls
- All comms via Orchestrator
- All comms via Orchestrator

SECURITY OBSERVER (Read-Only)

ZERO write scopes

- Watches all tool calls
- MITRE ATLAS threat tagging
- Prompt injection detection
- Supply-chain anomaly alerts

Observer Recommends | Policy Router enforces

WRITE SAFETY (TRANSACTION FW)

AGENTS
Propose



Firewall
Approve



SaaS
Execute

- ABAC policy enforcement
- > \$250 → Human approval
- Idempotency keys
- Step-up authentication

COMPLIANCE EVIDENCE

Bi-temporal Decision Trace

- What AI knew at decision time
- What evidence was retrieved
- What action was taken
- Policy version applied

Supports: ISO 42001, NIST AI RMF, EU AI Act,

RESILIENCE + IMPLEMENTATION

WHEN AI FAILS (Graceful Degradation)

"Every agent assumed compromised"

Agent Error → Auto-retry (3x) → Rule-based fallback → Assist mode → Human escalation

Low RAG confidence	→	Corrective RAG (broaden + verify)
Prompt injection	→	Block + alert Security Observer
High-risk decision	→	Route to Transaction Firewall HITL

IMPLEMENTATION PHASES

PHASE 1 (4wk)

Infra + Auth
3 VPCs + Link
Firewall MVP

PHASE 2 (4wk)

Integrations
RAG + Webhooks
RAGAS > 0.8

PHASE 3 (4wk)

Launch + Scale
Beta → Prod
< 5% human

TOGAF

Architecture Roadmap aligned to Business Capability increments

agentLUMEN

The slide features a large orange arrow pointing right, which serves as a background for the text. On the far left, there are three vertical bars: a thick orange one, a thin white one, and a thin dark grey one. The text 'agentLUMEN' is centered in the top orange box, with 'agent' in a lowercase, rounded font and 'LUMEN' in a bold, uppercase, sans-serif font. The 'L' is white, while the rest of the letters are black. Below this, the text 'NEXT STEPS' is centered in a bold, uppercase, sans-serif font. At the bottom, there is a large, empty orange rectangular box with a thin white border.

agentLUMEN

NEXT STEPS